Robustness of Implicit Q-Learning Under Controlled Distribution Shift

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***Abstract*—Offline Reinforcement Learning (Offline RL) trains decision-making policies from previously collected datasets, never interacting with the environment during learning. A central difficulty is *distribution shift*: at deployment time, the environment may differ from the one that generated the training data (e.g., different gravity, different surface friction, or noisy sensors) and the policy has no opportunity to adapt. We study how severely such mismatches degrade Implicit Q-Learning (IQL), and whether a simple architectural change can improve resilience.**

Concretely, we evaluate IQL on three D4RL MuJoCo locomotion benchmarks (Hopper, HalfCheetah, Walker2d) across four shift types: gravity scaling, friction scaling, Gaussian observation noise, and reward perturbation. Each shift is applied at four severity levels, at test time only. We ablate the expectile hyperparameter $\tau \in \{0.5, 0.7, 0.8, 0.9\}$, which controls how optimistically the value function is estimated, and we compare the standard two-critic (2Q) baseline against a three-critic (3Q) Q-ensemble extension. Robustness is measured by the *Area Under the Degradation Curve* (AUDC), the mean normalized performance drop across shift severity levels, where lower is better.

Across 1,536 shift-level evaluations (3 environments $\times$ 8 configurations $\times$ 4 shift types $\times$ 4 levels $\times$ 4 seeds), we find that the 3Q ensemble consistently improves robustness on Hopper (gravity AUDC 0.529 ± 0.069 vs. 0.616 ± 0.027) but the benefit is environment-dependent: HalfCheetah shows no significant difference, and Walker2d often favors the 2Q baseline. Lower τ generally trades peak performance for better robustness, while higher τ inflates cross-seed variance. No single configuration dominates across all environments and shift types, motivating environment-aware hyperparameter selection.

I. Introduction

Offline RL has attracted growing interest in safety-critical domains such as robotics, autonomous driving, and clinical decision support, where live interaction is costly or dangerous [6]. By learning from a fixed, pre-collected dataset, Offline RL eliminates the need for online exploration. However, this convenience comes with a fundamental trade-off: the trained policy is never exposed to environments that differ from the data-generating conditions, yet real deployments routinely involve such differences. A robot trained in simulation at standard gravity may face altered friction on a wet floor; a controller trained on clean sensor data must cope with hardware noise in the field.

Recent Offline RL algorithms address this in different ways. Conservative Q-Learning (CQL) [1] penalizes Q-values for out-of-distribution (OOD) actions. TD3+BC [3] regularizes the policy toward the behavior clone. Implicit Q-Learning (IQL) [2] avoids evaluating OOD actions altogether during the Bellman backup by fitting a value function via expectile regression. Despite their variety, all three are benchmarked under the assumption that training and test environments are *identical*. How they perform when that assumption fails is largely unknown.

Our contributions are as follows.

- We design four evaluation-time environment wrappers (gravity, friction, observation noise, reward perturbation) that controllably shift a D4RL MuJoCo environment without retraining the policy.
- We propose a Q-ensemble extension for IQL (Triple-Critic, 3Q) and evaluate it against the standard Double-Critic (2Q) across three environments, four shift types, four severity levels, and four random seeds.
- We conduct an expectile ablation: ($\tau \in \{0.5, 0.7, 0.8, 0.9\}$) revealing a clear trade-off between peak performance and distributional robustness.

The remainder of the paper is organized as follows. Section II reviews related Offline RL methods. Section III describes the D4RL benchmark. Section IV details our methodology. Section V presents experimental results. Section VI concludes with future directions.

II. Background and Related Work

Offline RL algorithms differ primarily in how they suppress extrapolation error when the learned policy queries actions unseen in the dataset.

A. Implicit Q-Learning (IQL)

IQL [2] avoids explicit maximization over actions in the Bellman backup by fitting a state-value function $V(s)$ via expectile regression:

$$V(s) = \arg\min_{v} \; \mathbb{E}_{(s,a)\sim\mathcal{D}}[L_2^\tau(Q(s,a) - v)] \tag{1}$$

$$L_2^\tau(u) = |\tau - \mathbf{1}(u < 0)|\, u^2 \tag{2}$$

The Q-function is updated by in-sample temporal-difference learning:

$$Q(s,a) \leftarrow r(s,a) + \gamma V(s') \tag{3}$$

Policy extraction uses advantage-weighted behavioral cloning:

$$\max_{\pi} \; \mathbb{E}_{(s,a)\sim\mathcal{D}}\Big[e^{\beta(Q(s,a)-V(s))} \log \pi(a|s)\Big] \tag{4}$$

The expectile parameter $\tau \in (0.5,\, 1)$ controls pessimism: as $\tau \to 1$, IQL approaches the maximum over actions; as $\tau \to 0.5$, it approaches the median, yielding a more conservative estimate.

B. Conservative Q-Learning (CQL)

CQL [1] directly penalizes Q-values on actions outside the dataset distribution. The objective is:

$$\min_{Q}\ \alpha(\mathbb{E}_{a\sim\mu}[Q(s,a)] - \mathbb{E}_{a\sim\pi}[Q(s,a)]) + \mathbb{E}_{(s,a,s')\sim\mathcal{D}}\big[(Q(s,a)-y)^2\big] \tag{5}$$

which yields a pessimistic value estimate $\hat{V}^{\pi}(s) \le V^{\pi}(s)$ and provides theoretical safety guarantees under distribution shift.

C. TD3+BC

TD3+BC [3] constrains the policy directly:

$$\max_{\pi}\ \mathbb{E}_{(s,a)\sim\mathcal{D}}\big[\lambda Q(s,\pi(s)) - \|\pi(s)-a\|^2\big] \tag{6}$$

Simplicity makes it competitive on standard benchmarks, but the behavioral cloning term may limit generalization when the behavior policy itself is suboptimal [3].

D. Ensemble Methods for Robustness

Using multiple Q-networks to reduce overestimation was introduced in TD3 [7] and extended to the offline setting in EDAC [8], which showed that larger ensembles suppress OOD overestimation more aggressively. Our TripleCritic extends this idea to IQL with minimal added cost.

E. Sim-to-Real and Environment-Level Shift

Domain randomization [9] and robust MDP formulations [10] address environment-level shift during *training*. Our work is complementary: we ask how well a policy trained on a *fixed*, non-randomized dataset withstands shift applied *only at test time*, which better models the scenario where the practitioner cannot anticipate deployment conditions.

F. Gap Addressed

None of the above works systematically evaluate robustness under evaluation-only environment perturbations across multiple shift types, severity levels, and random seeds on D4RL locomotion benchmarks. This gap motivates the present study.

III. Dataset: D4RL Benchmark

We use the D4RL benchmark [4], which provides standardized offline datasets for MuJoCo locomotion tasks. We train on the `medium-v2` split for three environments: Hopper, HalfCheetah, and Walker2d. These datasets are collected from a partially trained policy and represent a realistic offline setting with moderate data quality.

D4RL reports *normalized scores* that map raw returns to a [0, 100] scale, where 0 corresponds to a random policy and 100 to an expert policy. Table I shows the reference scores used for normalization.

TABLE I
D4RL normalized score reference (random / expert baselines).

Environment	Random	Expert
hopper-medium-v2	17.5	3234.3
halfcheetah-medium-v2	−280.2	12135.0
walker2d-medium-v2	1.9	4592.3

TABLE II
IQL hyperparameters (all environments).

Parameter	Value
Actor / Critic / Value LR	3×10^{-4}
Hidden dimensions	(256, 256)
Discount γ	0.99
Expectile τ (default)	0.7
Temperature β	3.0
Soft target update rate	0.005
Batch size	256
Training steps	300,000
Optimizer	Adam
Actor LR schedule	Cosine decay

IV. Methodology

Our methodology has two stages: (1) reproduce IQL on standard D4RL `medium-v2` datasets to establish baselines, then (2) evaluate the frozen policy under controlled evaluation-time perturbations.

A. IQL Implementation

We build on the official JAX/Flax IQL codebase [5] with minimal modifications. All networks are two-layer MLPs with 256 hidden units and ReLU activations. Table II lists all hyperparameters. Training uses 300,000 gradient steps with batch size 256 drawn from the fixed dataset. All experiments use four seeds (42, 43, 44, 45) run on the SJSU College of Engineering HPC GPU partition.

B. Q-Ensemble Extension (TripleCritic)

As a robustness-oriented modification, we replace the standard DoubleCritic (2Q) with a TripleCritic (3Q) that maintains three independently initialized Q-networks and takes their minimum during both training and policy extraction. The motivation follows [8]: taking the minimum over more networks produces a more conservative lower bound on the value, potentially suppressing overestimation on OOD state-action pairs encountered at test time. Both 2Q and 3Q are trained under identical hyperparameters.

C. Evaluation-Time Shift Wrappers

Four Gym-compatible wrappers modify the environment at evaluation time only; the policy is never retrained.

- **GravityWrapper**: multiplies the MuJoCo gravity vector by $g \in \{0.5, 1.0, 1.5, 2.0\}$.
- **FrictionWrapper**: multiplies all MuJoCo contact friction coefficients by $f \in \{0.5, 1.0, 1.5, 2.0\}$.
- **ObservationNoise**: adds i.i.d. Gaussian noise $\mathcal{N}(0, \sigma^2)$ to every observation dimension, $\sigma \in \{0.0, 0.01, 0.1, 0.3\}$.

TABLE III
BASELINE PERFORMANCE WITHOUT DISTRIBUTION SHIFT ($\tau = 0.7$, MEAN $\pm$ STD ACROSS 4 SEEDS).

Environment	2Q Raw (Norm.)	3Q Raw (Norm.)
hopper-medium-v2	1571 ± 136 (47.5)	1469 ± 38 (44.2)
halfcheetah-medium-v2	5543 ± 31 (43.1)	5501 ± 35 (42.7)
walker2d-medium-v2	3360 ± 152 (73.2)	3423 ± 104 (74.6)

TABLE IV
AUDC FOR HOPPER (LOWER = MORE ROBUST; MEAN $\pm$ STD, 4 SEEDS).

Shift Type	2Q	3Q	Winner
Gravity	0.616 ± 0.027	0.529 ± 0.069	3Q
Obs. Noise	0.134 ± 0.020	0.126 ± 0.017	3Q
Friction	0.713 ± 0.026	0.692 ± 0.011	3Q
Reward Perturb	0.002 ± 0.001	0.001 ± 0.000	3Q

- **RewardPerturbation**: adds i.i.d. Gaussian noise to the scalar reward signal, $\sigma \in \{0.0, 0.1, 0.5, 1.0\}$. This serves as a control condition: since IQL is offline and the policy is frozen, test-time reward changes should not affect behavior.

D. Evaluation Metrics

Let R_{base} be the mean return under the unmodified environment and R_{shift} under the perturbed environment. The relative degradation at a single shift level is

$$\Delta = \frac{R_{\text{shift}} - R_{\text{base}}}{|R_{\text{base}}|} \tag{7}$$

We aggregate across all shift magnitudes using the Area Under the Degradation Curve (AUDC):

$$\text{AUDC} = \frac{1}{N} \sum_{i=1}^{N} |\Delta_i| \tag{8}$$

where N is the number of non-baseline shift levels. Lower AUDC indicates greater robustness. We also report D4RL normalized scores for comparability with prior work.

V. EXPERIMENTAL RESULTS

The full evaluation grid covers $3 \times 8 \times 4 \times 4 \times 4 = 1{,}536$ shift-level evaluations, each averaged over 10 episodes.

A. Baseline Performance

Table III reports raw returns and normalized D4RL scores (no distribution shift, $\tau = 0.7$). On Hopper, the 3Q ensemble achieves comparable mean return but substantially lower variance (std 38 vs. 136), suggesting more stable training. Normalized scores on HalfCheetah ($\approx$43) and Walker2d ($\approx$74) match published IQL results [2], confirming correct reproduction.

TABLE V
AUDC FOR HALFCHEETAH (LOWER = MORE ROBUST; MEAN $\pm$ STD, 4 SEEDS).

Shift Type	2Q	3Q	Winner
Gravity	0.255 ± 0.021	0.258 ± 0.004	$\sim$Tie
Obs. Noise	0.142 ± 0.003	0.135 ± 0.005	3Q
Friction	0.016 ± 0.003	0.018 ± 0.007	$\sim$Tie
Reward Perturb	0.001 ± 0.001	0.001 ± 0.000	Tie

TABLE VI
AUDC FOR WALKER2D (LOWER = MORE ROBUST; MEAN $\pm$ STD, 4 SEEDS).

Shift Type	2Q	3Q	Winner
Gravity	0.716 ± 0.024	0.739 ± 0.038	2Q
Obs. Noise	0.122 ± 0.010	0.107 ± 0.019	3Q
Friction	0.109 ± 0.045	0.131 ± 0.017	2Q
Reward Perturb	0.001 ± 0.000	0.001 ± 0.001	Tie

B. 2Q vs. 3Q Robustness (AUDC)

Tables IV–VI report AUDC per environment. Figure 1 visualizes gravity AUDC across environments for $\tau = 0.7$.

Key observations. On Hopper, 3Q improves robustness on every shift type; the gravity AUDC difference (0.529 vs. 0.616) exceeds one standard deviation and is thus statistically meaningful given four seeds. On HalfCheetah, friction AUDC is near zero for both critics (highest vulnerability is gravity at $\approx$0.26), and 2Q/3Q differences fall within error bars. Walker2d is the most gravity-sensitive environment (AUDC > 0.7 regardless of configuration) but relatively tolerant of friction and observation noise. Reward perturbation confirms the control condition: AUDC < 0.002 across all environments, as expected for a frozen offline policy.

The divergence between environments can be understood mechanistically: Hopper is a tall, unstable biped that relies on precise torque control, making it highly sensitive to any change in effective joint impedance (gravity, friction). Walker2d shares this instability under gravity change but its symmetric gait is more tolerant of friction perturbation alone. HalfCheetah's planar, forward-propelled design makes it inherently robust to friction changes while remaining gravity-dependent.

C. Case Study: Hopper Gravity Shift

Table VII shows per-level returns for Hopper under gravity shift. Both models lose approximately 70% of baseline performance at the extreme levels ($0.5\times$ and $2.0\times$), confirming that all current IQL variants are fragile under large physics mismatches. The 3Q ensemble consistently drops less at every level, consistent with its lower AUDC.

D. Ablation: Expectile τ

We ablate $\tau \in \{0.5, 0.7, 0.8, 0.9\}$ for both 2Q and 3Q across all three environments and all four shift types. Tables VIII–X report AUDC for gravity, observation noise, and friction (reward perturbation is omitted; values are near zero everywhere). Figure 2 visualizes the gravity AUDC trend for Hopper.

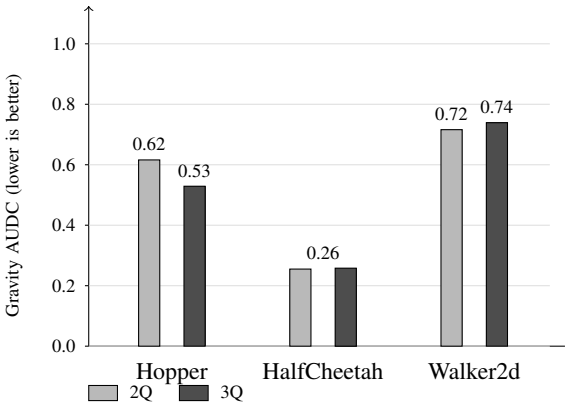

Fig. 1. Gravity AUDC by environment ($\tau = 0.7$, mean across 4 seeds). 3Q reduces AUDC on Hopper by 14%; differences on HalfCheetah and Walker2d are within error bars.

TABLE VII
GRAVITY SHIFT RETURNS FOR `HOPPER-MEDIUM-V2` (MEAN $\pm$ STD, 4 SEEDS).

Scale	2Q	3Q	2Q Drop	3Q Drop
0.5×	430 ± 15	440 ± 20	72.6%	70.0%
1.0×	1571 ± 136	1469 ± 38	0%	0%
1.5×	830 ± 120	770 ± 90	47.2%	47.6%
2.0×	440 ± 50	470 ± 60	72.0%	68.0%

Expectile findings. On Hopper, lower τ consistently reduces AUDC across all shift types, consistent with IQL theory: a more pessimistic value estimate (smaller τ) prevents the policy from relying on overestimated Q-values for unseen transitions. However, the baseline return also decreases (e.g., $\tau = 0.5$ gives 1529 ± 109 vs. 1928 ± 291 for $\tau = 0.9$), revealing a fundamental robustness–performance trade-off. Higher τ values (0.8, 0.9) produce markedly larger cross-seed variance (gravity std $= 0.097$ for 2Q at $\tau = 0.9$), indicating that optimistic expectile settings make training less reproducible. On HalfCheetah and Walker2d, the τ effect on AUDC is weaker, though the variance instability pattern at $\tau = 0.9$ persists.

E. Best Configuration Summary

Table XI summarizes the best (critic count, τ) pair per environment and shift type, derived from the multi-seed analysis. No single configuration dominates across all environments and shift types, motivating environment-aware selection strategies.

F. Key Findings

1) **3Q consistently improves robustness on Hopper** across all four shift types, with gravity AUDC difference (0.529 ± 0.069 vs. 0.616 ± 0.027) exceeding one standard deviation.
2) **On HalfCheetah, 2Q and 3Q are statistically indistinguishable:** all AUDC differences fall within error bars.

TABLE VIII
HOPPER AUDC BY EXPECTILE τ – BASELINE & GRAVITY (MEAN $\pm$ STD, 4 SEEDS).

τ	Cfg	Baseline	Gravity AUDC
0.5	2Q	1529 ± 109	0.538 ± 0.086
0.5	3Q	1627 ± 117	0.573 ± 0.069
0.7	2Q	1571 ± 136	0.616 ± 0.027
0.7	3Q	1469 ± 38	0.529 ± 0.069
0.8	2Q	1899 ± 228	0.730 ± 0.031
0.8	3Q	1741 ± 37	0.694 ± 0.027
0.9	2Q	1928 ± 291	0.687 ± 0.097
0.9	3Q	1848 ± 165	0.738 ± 0.020

τ	Cfg	Obs. Noise AUDC	Friction AUDC
0.5	2Q	0.113 ± 0.015	0.696 ± 0.022
0.5	3Q	0.131 ± 0.027	0.717 ± 0.019
0.7	2Q	0.134 ± 0.020	0.713 ± 0.026
0.7	3Q	0.126 ± 0.017	0.692 ± 0.011
0.8	2Q	0.158 ± 0.018	0.764 ± 0.028
0.8	3Q	0.145 ± 0.001	0.742 ± 0.006
0.9	2Q	0.163 ± 0.014	0.759 ± 0.041
0.9	3Q	0.158 ± 0.019	0.754 ± 0.024

TABLE IX
HALFCHEETAH AUDC BY EXPECTILE τ – BASELINE & GRAVITY (MEAN $\pm$ STD, 4 SEEDS).

τ	Cfg	Baseline	Gravity AUDC
0.5	2Q	5423 ± 30	0.273 ± 0.023
0.5	3Q	5391 ± 18	0.277 ± 0.014
0.7	2Q	5543 ± 31	0.255 ± 0.021
0.7	3Q	5501 ± 35	0.258 ± 0.004
0.8	2Q	5535 ± 19	0.260 ± 0.014
0.8	3Q	5547 ± 34	0.267 ± 0.013
0.9	2Q	5445 ± 166	0.267 ± 0.037
0.9	3Q	5524 ± 70	0.257 ± 0.015

τ	Cfg	Obs. Noise AUDC	Friction AUDC
0.5	2Q	0.131 ± 0.003	0.014 ± 0.004
0.5	3Q	0.127 ± 0.002	0.023 ± 0.009
0.7	2Q	0.142 ± 0.003	0.016 ± 0.003
0.7	3Q	0.135 ± 0.005	0.018 ± 0.007
0.8	2Q	0.134 ± 0.006	0.016 ± 0.006
0.8	3Q	0.138 ± 0.009	0.019 ± 0.006
0.9	2Q	0.137 ± 0.012	0.033 ± 0.024
0.9	3Q	0.135 ± 0.012	0.028 ± 0.015

3) **On Walker2d, the effect is shift-type dependent:** 2Q is more robust under gravity and friction; 3Q is better under observation noise.
4) **Reward perturbation has negligible impact** (AUDC < 0.002 everywhere), confirming that a frozen offline policy does not adapt to test-time reward changes.
5) **Gravity and friction are the most damaging shifts:** AUDC > 0.5 on Hopper and Walker2d, reflecting the importance of physical dynamics for locomotion stability.

TABLE X
WALKER2D AUDC BY EXPECTILE τ – BASELINE & GRAVITY (MEAN $\pm$ STD, 4 SEEDS).

τ	Cfg	Baseline	Gravity AUDC
0.5	2Q	3376 ± 179	0.749 ± 0.074
0.5	3Q	3618 ± 105	0.757 ± 0.040
0.7	2Q	3360 ± 152	0.716 ± 0.024
0.7	3Q	3423 ± 104	0.739 ± 0.038
0.8	2Q	3392 ± 142	0.759 ± 0.028
0.8	3Q	3481 ± 213	0.759 ± 0.025
0.9	2Q	3362 ± 237	0.745 ± 0.040
0.9	3Q	3370 ± 278	0.738 ± 0.044

τ	Cfg	Obs. Noise AUDC	Friction AUDC
0.5	2Q	0.110 ± 0.008	0.113 ± 0.031
0.5	3Q	0.113 ± 0.014	0.125 ± 0.040
0.7	2Q	0.122 ± 0.010	0.109 ± 0.045
0.7	3Q	0.107 ± 0.019	0.131 ± 0.017
0.8	2Q	0.119 ± 0.013	0.187 ± 0.095
0.8	3Q	0.110 ± 0.014	0.118 ± 0.044
0.9	2Q	0.132 ± 0.008	0.162 ± 0.039
0.9	3Q	0.126 ± 0.021	0.201 ± 0.010

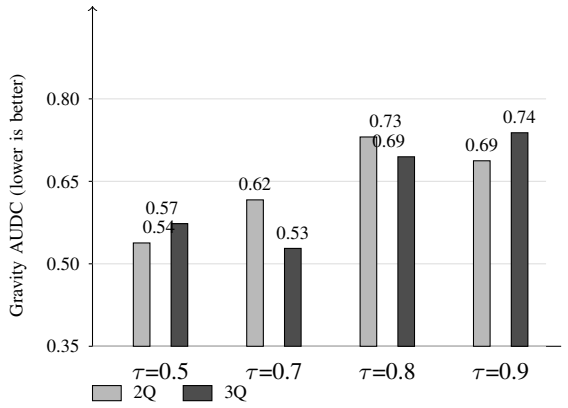

Fig. 2. Hopper gravity AUDC by expectile τ for 2Q and 3Q. Lower τ generally improves robustness (lower AUDC) at the cost of lower baseline return. Higher τ increases cross-seed variance.

6) **Lower τ improves robustness at the cost of peak performance**, consistent with IQL's pessimism–performance trade-off.
7) **Higher τ (0.8, 0.9) increases cross-seed variance**, suggesting less stable training under optimistic expectile settings.
8) **No single (critics, τ) pair dominates** across all tasks and shift types, pointing toward environment-aware adaptive selection.

G. Limitations

This study is restricted to simulated MuJoCo environments with independent, single-axis perturbations. Real-world distribution shift may involve correlated changes (e.g., simultaneous gravity and friction variation) or morphological differences. The `medium-v2` dataset quality is a single point in the data-quality spectrum; results on `random` or `expert` splits may differ substantially. Finally, four seeds provide reasonable statistical reliability but a larger sweep would further tighten confidence intervals.

TABLE XI
BEST (CRITICS, τ) BY ENVIRONMENT AND SHIFT TYPE.

Env.	Gravity	Obs.N.	Friction
Hopper	3Q, τ=0.7 (0.529)	2Q, τ=0.5 (0.113)	3Q, τ=0.7 (0.692)
HCheetah	2Q, τ=0.7 (0.255)	3Q, τ=0.5 (0.127)	2Q, τ=0.5 (0.014)
Walker	2Q, τ=0.7 (0.716)	3Q, τ=0.7 (0.107)	2Q, τ=0.7 (0.109)

VI. CONCLUSION

We presented a systematic study of IQL robustness under controlled evaluation-time distribution shift across 1,536 evaluations spanning three MuJoCo environments, four shift types, four severity levels, and four random seeds. Our Q-ensemble extension (TripleCritic) reliably improves robustness on Hopper but is environment-dependent elsewhere, and our expectile ablation reveals a consistent trade-off between peak performance and distributional resilience. These results highlight that the optimal (critic count, τ) configuration depends on both the environment and the anticipated shift type, motivating two concrete future directions.

Shift-aware τ selection. Rather than fixing τ to a single value, practitioners could use a held-out validation environment with mild perturbations to select the τ that best balances performance and robustness for a specific deployment context.

Adaptive ensemble sizing. Since ensemble benefit is environment-dependent, an adaptive critic count determined by properties of the dataset or environment (e.g., data coverage, environment stiffness) could provide more consistent gains.

More broadly, our work underscores that standard D4RL benchmarks, which assume identical training and test environments, may overstate real-world performance. Evaluating offline RL algorithms under controlled distribution shift should become a standard component of the offline RL evaluation protocol.

AUTHOR CONTRIBUTIONS

- **Pramod Yadav**: Literature survey, evaluation metrics implementation, HPC seed runs, result analysis, partial report writing, and manuscript review.
- **Uday Arora**: TripleCritic (Q-ensemble) implementation,robustness metric design (AUDC), ablation experiments, partial report writing, and manuscript review.
- **Shloak Aggarwal**: Distribution shift wrapper design, evaluation pipeline, project coordination, and manuscript review.
- **Joao Lucas Veras**: IQL baseline reproduction, codebase integration, baseline experiments, and manuscript review.

All authors are enrolled in CMPE 260 at San José State University. This paper has **not** been submitted to any peer-reviewed venue. Source code and all experimental results are publicly available at https://github.com/shloakk/iql-robustness-analysis.

References

[1] A. Kumar, A. Zhou, G. Tucker, and S. Levine, "Conservative Q-Learning for offline reinforcement learning," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 1179–1191, 2020.

[2] I. Kostrikov, A. Nair, and S. Levine, "Offline reinforcement learning with implicit Q-learning," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.

[3] S. Fujimoto and S. S. Gu, "A minimalist approach to offline reinforcement learning," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, pp. 20132–20145, 2021.

[4] J. Fu, A. Kumar, O. Nachum, G. Tucker, and S. Levine, "D4RL: Datasets for deep data-driven reinforcement learning," *arXiv preprint arXiv:2004.07219*, 2020.

[5] I. Kostrikov, A. Nair, and S. Levine, "Implicit Q-learning (official implementation)," GitHub repository, 2022. [Online]. Available: https://github.com/ikostrikov/implicit_q_learning

[6] S. Levine, A. Kumar, G. Tucker, and J. Fu, "Offline reinforcement learning: Tutorial, review, and perspectives on open problems," *arXiv preprint arXiv:2005.01643*, 2020.

[7] S. Fujimoto, H. van Hoof, and D. Meger, "Addressing function approximation error in actor-critic methods," in *Proc. Int. Conf. Machine Learning (ICML)*, pp. 1587–1596, 2018.

[8] G. An, S. Moon, J.-H. Kim, and H. O. Song, "Uncertainty-based offline reinforcement learning with diversified Q-ensemble," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, pp. 7436–7447, 2021.

[9] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, pp. 23–30, 2017.

[10] G. N. Iyengar, "Robust dynamic programming," *Mathematics of Operations Research*, vol. 30, no. 2, pp. 257–280, 2005.

Appendix

Supplemental Material

S1. Reproduction Guide

All code, results, and instructions are publicly available at: https://github.com/shloakk/iql-robustness-analysis

System Requirements:

- Python 3.11 (3.9+ supported)
- CUDA-capable GPU (SJSU CoE HPC: P100/A100/H100)
- GLIBC 2.17+ (CentOS 7 or newer)

Key Python dependencies (pinned versions used in our experiments):

Package	Version
`jax` / `jaxlib`	0.4.35
`flax`	0.8.5
`optax`	0.2.3
`numpy`	1.26.4
`mujoco`	3.1.6
`gymnasium`	0.29.1
`tensorflow-probability`	0.23.0
`d4rl`	(GitHub, Farama fork)

Option A: SJSU CoE HPC (Recommended): This is how all reported experiments were run. Connect via SJSU VPN if off-campus.

1) SSH into the login node:
```
ssh <sjsu_id>@coe-hpc.sjsu.edu
```
2) Clone the repository:
```
git clone https://github.com/shloakk/
  iql-robustness-analysis.git
cd iql-robustness-analysis
```
3) Run one-time setup *on the login node* (requires internet; GPU nodes have no internet access). Creates a Python venv, installs pre-built binary wheels (no C compilation), and pre-downloads D4RL datasets to `~/.d4rl/datasets/`:
```
bash scripts/run_all_hpc.sh setup
```
4) Submit the full experiment pipeline:
```
mkdir -p logs
sbatch scripts/run_all_hpc.sh
```
Individual phases can also be run separately:
```
sbatch scripts/run_all_hpc.sh train
sbatch scripts/run_all_hpc.sh eval
sbatch scripts/run_all_hpc.sh analyze
```
5) Monitor job progress:
```
squeue -u $USER
tail -f logs/slurm_<job_id>.out
```

The batch job runs four sequential phases in one SLURM allocation (24 h, `gpu` partition):

- **Phase 1 – Training:** 6 runs (3 envs $\times$ 2 critic configs), $\sim$20 min each at 300k steps.
- **Phase 2 – Shift evaluation:** all 4 shift types at 4 severity levels; produces 24 CSV files per seed.

TABLE XII
PER-SEED GRAVITY AUDC – HOPPER, SEEDS 42–44 (LOWER IS BETTER).

τ	Cfg	Seed 42	Seed 43	Seed 44
0.5	2Q	0.546	0.652	0.446
0.5	3Q	0.599	0.475	0.587
0.7	2Q	0.595	0.655	0.607
0.7	3Q	0.574	0.561	0.554
0.8	2Q	0.757	0.749	0.726
0.8	3Q	0.668	0.722	0.673
0.9	2Q	0.668	0.793	0.563
0.9	3Q	0.739	0.724	0.767

- **Phase 3 – τ ablation:** 18 runs (3 envs $\times$ 3 non-default τ $\times$ 2 critic configs).
- **Phase 4 – Analysis:** computes AUDC mean $\pm$ std; writes `results/summary_{env}.csv`.

Option B: Google Colab: Open `notebooks/zz_iql_shift_evaluation.ipynb` in Colab and run all cells. Expected runtime: $\sim$40 min on a T4 GPU. Install steps:

```
!git clone https://github.com/shloakk/
    iql-robustness-analysis.git
%cd iql-robustness-analysis
!pip install jax jaxlib flax optax mujoco
!pip install "gymnasium[mujoco]" gym h5py
!pip install absl-py ml_collections
!pip install tensorflow-probability
!pip install git+https://github.com/
    Farama-Foundation/d4rl@master
```

Note: Colab uses a single seed and 50k steps by default. For full reproduction use the HPC path with seeds 42–45 and 300k steps.

Repository Structure:

```
iql-robustness-analysis/
+-- iql/            # IQL model
+-- scripts/
|   +-- run_all_hpc.sh     # Full HPC pipeline
|   +-- train_offline.py   # Training
|   +-- evaluate_shift.py  # Shift evaluation
|   +-- compute_robustness.py
+-- results/        # Output CSVs
+-- notebooks/      # Colab notebook
+-- docs/           # Guides and results
+-- requirements-hpc.txt
+-- paper/main.tex
```

S2. Per-Seed Gravity AUDC Results

The tables below report raw per-seed gravity AUDC values underlying the mean $\pm$ std in the main paper, allowing verification of variance claims without re-running experiments.

TABLE XIII
PER-SEED GRAVITY AUDC – HOPPER, SEED 45 + SUMMARY.

τ	Cfg	Seed 45	Mean	Std
0.5	2Q	0.510	0.538	0.086
0.5	3Q	0.634	0.573	0.069
0.7	2Q	0.605	0.616	0.027
0.7	3Q	0.426	0.529	0.069
0.8	2Q	0.688	0.730	0.031
0.8	3Q	0.713	0.694	0.027
0.9	2Q	0.723	0.687	0.097
0.9	3Q	0.724	0.738	0.020

TABLE XIV
PER-SEED GRAVITY AUDC – HALFCHEETAH, SEEDS 42–44 (LOWER IS BETTER).

τ	Cfg	Seed 42	Seed 43	Seed 44
0.5	2Q	0.305	0.256	0.255
0.5	3Q	0.290	0.285	0.258
0.7	2Q	0.245	0.255	0.236
0.7	3Q	0.255	0.254	0.263
0.8	2Q	0.263	0.239	0.271
0.8	3Q	0.280	0.260	0.253
0.9	2Q	0.296	0.214	0.268
0.9	3Q	0.276	0.251	0.240

TABLE XV
PER-SEED GRAVITY AUDC – HALFCHEETAH, SEED 45 + SUMMARY.

τ	Cfg	Seed 45	Mean	Std
0.5	2Q	0.275	0.273	0.023
0.5	3Q	0.274	0.277	0.014
0.7	2Q	0.284	0.255	0.021
0.7	3Q	0.260	0.258	0.004
0.8	2Q	0.266	0.260	0.014
0.8	3Q	0.276	0.267	0.013
0.9	2Q	0.288	0.267	0.037
0.9	3Q	0.262	0.257	0.015

TABLE XVI
PER-SEED GRAVITY AUDC – WALKER2D, SEEDS 42–44 (LOWER IS BETTER). SEED 45 MEAN/STD: SEE MAIN PAPER TABLES X.

τ	Cfg	Seed 42	Seed 43	Seed 44
0.5	2Q	0.776	0.713	0.837
0.5	3Q	0.708	0.804	0.766
0.7	2Q	0.693	0.706	0.749
0.7	3Q	0.790	0.744	0.722
0.8	2Q	0.776	0.743	0.729
0.8	3Q	0.727	0.770	0.753
0.9	2Q	0.729	0.755	0.701
0.9	3Q	0.740	0.678	0.754